

FIE401 - Assignment

M&A and Method of Payment

Darya Yuferova

Overview

Research question: Do mergers destroy value for bidder shareholders depending on the method of payment?

Corporate takeovers are among the largest investments that a company will ever undertake. M&A deals provide researchers with a unique opportunity to analyze value implications of important managerial decisions. Empirical research in this area has focused on a wide range of topics including size and division of takeovers gains (short- and long- run abnormal stock returns to bidders and targets). Data availability has allowed researchers to conduct empirical tests based on unprecedented sample sizes and detail.

In this assignment, you have to evaluate the announcement effect of takeovers on the wealth of bidder shareholders and study the impact of the method of payment (stock versus cash). In addition, you ask whether this impact is different for public and private targets.

For a list of papers investigating the choice of payment in a merger setting, see Betton et al., 2008, Table 4.

Formalities

This assignment will be handed out on 24.02.2026 at 12:00 and has to be submitted no later than the **03.03.2026 at 09:00**. Submit your commented coding file (file extension: “.R”) and a pdf file (file extension: “.pdf”) including the numerical results and answers to questions posted.

Please comment your code shortly so that a reader can reconstruct your thinking. You do not need to explain the used functions. You do not need to describe your coding in the pdf file. Please keep your answers very brief. **Code without comments will automatically result in a "Fail" grade.**

On Canvas, you will find a guideline of how to make good tables. Note that in the last lecture of this course, we will cover the subject of how to present research in detail.

After submission, the assignments will be randomly redistributed. Read your peer’s work carefully and comment within Canvas no later than **10.03.2026 at 09:00**. Your comments are meant to improve the work of your peers, so use constructive feedback. See peer-review guidelines for details. **Insufficient quality of peer-review will result in a warning for Assignment 1. No peer-review will automatically result in a "Fail" grade, regardless of the quality of your own analysis.**

In order to make it easy for us to allocate individual assignment submission as well as peer review to groups, please follow following file name structure. **For the assignment submission: "Assignment1_group_XX.R/pdf"**. “XX” indicates the group number that submitted the assignment. **Please submit the peer review as a comment on CANVAS.**

Please do not include any personal information such as name, student number, etc. in the submitted files.

Note that in order to complete the assignment, you need to submit your own work as well as complete the peer review. Both aspects will be reviewed.

Work together in groups of 3-4 people.

Submit your assignment even if you do not finish all tasks.

Note that some concepts are introduced for the first time in this assignment. Please use online resources actively.

We touched on almost all functions you have to use to solve this assignment. If you are unsure how to use a function, either use R's own documentation (type `?command`) or use www.stackoverflow.com. We encourage you to code this assignment yourselves, and not to use purpose-made solutions or packages as provided on the internet.

Before starting to code, try to separate the task into smaller pieces.

Data provided for this assignment

CAR_M&A.Rdata contains 5,154 deals over the 1990-2014 period. The sample consists only of deals that fulfill a set of criteria that are typically used in the literature:

- Domestic transactions (U.S. acquirers and U.S. targets)
- Completed control transactions (acquirer holds less than 50% of the target shares before the announcement and ends up with 100% of the target shares)
- Public acquirers and targets of all statuses (private, public, subsidiaries)
- Deal value of at least 1 US\$ million
- Relative transaction size (ratio of the deal value to the acquirer market value) of at least 1%
- Financial industries (SIC codes 6000-6999) excluded
- Necessary information available in the CRSP and COMPUSTAT databases to compute the acquirer cumulative abnormal return (CAR) and the set of control variables.

The dataset includes a set of variables widely used in M&A literature (e.g., Moeller et al. (2004); Golubov et al. (2015)):

- `yyyymmdd`: year, month, and day of the deal announcement;
- `yyyy`: year of the deal announcement;
- `carbidd`: The acquirer CAR calculated over a three-day event window centered on the deal announcement, as reported in the SDC database. Abnormal returns are calculated using the market model. Estimation window is from day -300 to day -91 relative to the announcement on day 0;
- `deal_value`: the deal value in millions of U.S. dollars;
- `bidder_size`: the bidder's market value at the end of the fiscal year before the acquisition announcement in millions of U.S. dollars;
- `all_stock`: a dummy variable equal to 1 if the transaction is fully paid in stock;
- `private`: a dummy variable equal to 1 if the target is a private company;
- `public`: a dummy variable equal to 1 if the target is a public company;
- `bidder_mtb`: the acquirer market value of assets (defined as the book value of total assets minus common equity plus the market value of equity) divided by the acquirer book value of assets;
- `run_up_bidder`: the market-adjusted buy and hold return of the acquirer's stock price from day -210 to day -11 with respect to the announcement date;
- `bidder_fcf` (free cash-flow): the acquirer's operating income before depreciation minus interest expense and income taxes plus changes in deferred taxes and investment tax credit minus dividends on both preferred and common share divided by the book value of total assets;
- `bidder_lev` (leverage): the acquirer's long-term debt divided by the market value of assets, defined as above;
- `sigma_bidder`: the standard deviation of the acquirer market-adjusted daily returns from day -210 to day -11 with respect to the announcement date;
- `relsize`: the ratio of the deal value to the acquirer market value;
- `horz` (horizontal): a dummy variable equal to 1 if the bidder and the target operate in the same industry at the two-digit SIC code level;

- tender_offer: a dummy variable equal to 1 if the deal is classified as a tender offer in the SDC database; and
- hostile: a dummy equal to 1 if the transaction is classified as hostile in the SDC database.

Tasks

- Familiarize yourself with the data
 - Are there outliers? Evaluate if you should winsorize the data.
 - Are there some mistakes in the data? Evaluate to make adjustments to the data.
- Descriptive statistics (Table 1)
 - For each year in the sample, report the average deal size, bidder CAR, share of deals with private targets, and share of deals fully paid in stock.
- Descriptive statistics (Table 2)
 - By methods of payment (fully paid by stock or not), report the average of all variables you include in your regression model. Report the difference in means for both sub-samples and the associated p -value (use t -test).
- Regression (Table 3)
 - Make a regression table with 6 models.
 - The dependent variable is the bidder CAR.
 - The main independent variable is method of payment and whether or not the target is public or private.
 - Consider three sub-sample (2 models each): only deals with public target, only deals with private targets, and both.
 - For each sub-sample, do models with and without controls (3 times 2 = 6 models).
 - For the models with all deals, include an interaction term between method of payment and public/private status of target.
- Interpretation and additional questions
 - In writing, discuss the the insights from both descriptive tables for your research design.
 - In writing, discuss your regression results briefly (5-10 sentences). Make sure that you discuss statistic as well as economic significance. Regarding the latter, argue whether or not the detected effects are meaningful.
 - Answer following questions specifically [difficult]:
 - * Can you include both private and public in the regression? If not, why?
 - * If you limit the sample to private deals, what's the coefficient on the hostile variable?

References

- Betton, S., Eckbo, E., Thorburn, K., 2008, Corporate takeovers. In Handbook of Corporate Finance: Empirical Corporate Finance, Vol. 2, E. Eckbo, ed. Amsterdam: Elsevier/North- Holland
- Golubov, A., Yawson, A., Zhang, H., 2015, Extraordinary acquirers. Journal of Financial Economics, 116(2), pp. 314-330.
- Moeller, S., Schlingemann, Fr., Stulz, R., 2004, Firm size and the gains from acquisitions. Journal of Financial Economics, 73(2), pp. 201-228

```
#####
#                                     #
# Suggested code solution for Assignment 1 #
#                                     #
#####

# load necessary packages
require(DescTools)
require(dplyr)
require(tidyverse)
require(stargazer)
require(knitr)

# Set working directory
setwd("C:/Users/s13163/Dropbox/FIE401/data/data_assignments")

# Load data
load("CAR_M&A.Rdata")

# First look at the data
summary(CAR_MA)
```

```
##      yyyymmdd      yyyy      deal_value      public
## Min. :19900111 Min. :1990 Min. :      1.00 Min. :0.0000
## 1st Qu.:19960946 1st Qu.:1996 1st Qu.:     19.59 1st Qu.:0.0000
## Median :19991218 Median :1999 Median :      73.11 Median :0.0000
## Mean :20008863 Mean :2001 Mean :     835.17 Mean :0.4627
## 3rd Qu.:20051020 3rd Qu.:2005 3rd Qu.:    341.44 3rd Qu.:1.0000
## Max. :20141222 Max. :2014 Max. :164746.86 Max. :1.0000
##      private      tender_offer      all_stock      carbidder
## Min. :0.0000 Min. :0.000 Min. :0.0000 Min. : -0.590018
## 1st Qu.:0.0000 1st Qu.:0.000 1st Qu.:0.0000 1st Qu.: -0.042268
## Median :1.0000 Median :0.000 Median :0.0000 Median : 0.004455
## Mean :0.5373 Mean :0.111 Mean :0.3877 Mean : 0.009470
## 3rd Qu.:1.0000 3rd Qu.:0.000 3rd Qu.:1.0000 3rd Qu.: 0.054521
## Max. :1.0000 Max. :1.000 Max. :1.0000 Max. : 1.368253
##      bidder_size      sigma_bidder      run_up_bidder      hostile
## Min. :      400 Min. :0.006756 Min. :0.2385 Min. :0.000000
## 1st Qu.: 144882 1st Qu.:0.022137 1st Qu.:0.8204 1st Qu.:0.000000
## Median : 602535 Median :0.031337 Median :1.0404 Median :0.000000
## Mean : 5154525 Mean :0.036893 Mean :1.1908 Mean :0.008343
## 3rd Qu.: 2388443 3rd Qu.:0.044950 3rd Qu.:1.3323 3rd Qu.:0.000000
## Max. :432000000 Max. :0.279690 Max. :4.6878 Max. :1.000000
##      horz      relsize      bidder_mtb      bidder_fcf
## Min. :0.0000 Min. : 0.01001 Min. : 0.3235 Min. : -1.19328
## 1st Qu.:0.0000 1st Qu.: 0.04747 1st Qu.: 1.3867 1st Qu.: -0.01106
## Median :0.0000 Median : 0.12997 Median : 1.9175 Median : 0.04479
## Mean :0.3686 Mean : 0.37297 Mean : 2.8673 Mean : -0.01109
## 3rd Qu.:1.0000 3rd Qu.: 0.37102 3rd Qu.: 3.0703 3rd Qu.: 0.08605
## Max. :1.0000 Max. :12.17000 Max. :58.0409 Max. : 0.25274
##      bidder_lev
## Min. :0.000000
## 1st Qu.:0.001754
```

```
## Median :0.096706
## Mean   :0.154961
## 3rd Qu.:0.260215
## Max.   :0.746767

# Winsorize continuous variables at 99%
for(cn in c("deal_value","carbidder","bidder_size",
            "sigma_bidder","run_up_bidder","relnsize",
            "bidder_mtb","bidder_fcf","bidder_lev")){
  CAR_MA[[cn]] <- Winsorize(CAR_MA[[cn]], val=quantile(CAR_MA[[cn]],probs=c(0.005, 0.995)))
}

# Descriptives table 1: Generate summary statistics per year
# Note, I use tidyverse syntax
CAR_MA %>%
  group_by(yyyy) %>%
  summarise(N = n(), # observations
            mean_dealvalue = mean(deal_value), # average deal value
            share_private = mean(private), # share private targets
            all_stock = mean(all_stock), # share paid in stock
            mean_car = mean(carbidder) # bidder CAR
  ) %>%
  round(2) %>%
  kable() # nice table within Rmarkdown which I use to code your assignments
```

yyyy	N	mean_dealvalue	share_private	all_stock	mean_car
1990	67	200.52	0.45	0.48	-0.01
1991	103	100.66	0.61	0.51	0.02
1992	139	108.12	0.72	0.60	0.03
1993	171	198.51	0.67	0.49	0.04
1994	234	278.83	0.59	0.56	0.02
1995	320	360.73	0.58	0.58	0.01
1996	338	418.04	0.58	0.59	0.02
1997	398	296.21	0.54	0.54	0.02
1998	430	606.07	0.50	0.50	0.01
1999	391	963.88	0.47	0.52	0.01
2000	343	1160.94	0.54	0.53	-0.02
2001	216	793.71	0.39	0.39	-0.02
2002	169	385.62	0.50	0.28	-0.01
2003	171	401.48	0.49	0.26	0.02
2004	198	669.65	0.55	0.18	0.00
2005	217	1131.85	0.57	0.17	0.00
2006	192	985.86	0.51	0.16	0.01
2007	206	945.20	0.51	0.10	0.00
2008	127	780.92	0.60	0.17	0.00
2009	120	1198.68	0.45	0.20	0.00
2010	119	1005.39	0.44	0.12	0.01
2011	116	1360.55	0.61	0.09	0.00
2012	116	961.21	0.49	0.07	0.03
2013	107	920.55	0.57	0.11	0.03
2014	146	1728.40	0.56	0.17	0.02

```
# xtable(., type = "latex") # This code you can copy in a latex file

# Descriptives table 2:
# By methods of payment (fully paid by stock or not), report the average
# of all variables you include in your model.
# Report the difference in means for both sub-samples and the associated p-value
sapply(colnames(CAR_MA)[c(-1, -2, -7)], function(cn){
  t.test(CAR_MA[[cn]][CAR_MA$all_stock == 1],
        CAR_MA[[cn]][CAR_MA$all_stock == 0])[c("estimate", "p.value")] %>%
  unlist()
}) %>%
round(2) %>%
t() %>%
kable()
```

	estimate.mean of x	estimate.mean of y	p.value
deal_value	546.40	794.33	0.00
public	0.44	0.47	0.04
private	0.56	0.53	0.04
tender_offer	0.02	0.17	0.00
carbidder	0.01	0.01	0.32
bidder_size	3826481.77	5568237.36	0.00
sigma_bidder	0.04	0.03	0.00
run_up_bidder	1.31	1.12	0.00
hostile	0.00	0.01	0.00
horz	0.36	0.37	0.36
relsize	0.37	0.35	0.30
bidder_mtb	3.66	2.25	0.00
bidder_fcf	-0.04	0.01	0.00
bidder_lev	0.12	0.18	0.00

```
# Regressions

# Regressions for sub-sample of private targets
fit.1 <- lm(carbidder ~ all_stock,
            data = filter(CAR_MA, public == 1))

fit.2 <- lm(carbidder ~ all_stock
            + deal_value + bidder_size + bidder_lev
            + bidder_fcf + horz + hostile,
            data = filter(CAR_MA, public == 1))

# Regressions for sub-sample of private targets
fit.3 <- lm(carbidder ~ all_stock,
            data = filter(CAR_MA, private == 1))

fit.4 <- lm(carbidder ~ all_stock
            + log(deal_value) + log(bidder_size) + bidder_lev
            + bidder_fcf + horz + hostile,
            data = filter(CAR_MA, private == 1))
```

```
# Regressions for full sample with interatice
```

```
fit.5 <- lm(carbidder ~ all_stock * public,  
            data = CAR_MA)
```

```
fit.6 <- lm(carbidder ~ all_stock * public  
            + log(deal_value) + log(bidder_size) + bidder_lev  
            + bidder_fcf + horz + hostile,  
            data = CAR_MA)
```

```
# regression output
```

```
stargazer(  
  list(fit.1, fit.2, fit.3, fit.4, fit.5, fit.6),  
  type="text",  
  keep.stat=c("n", "rsq", "adj.rsq"),  
  report=('vc*t'))
```

```
##  
## =====  
##                               Dependent variable:  
##                               -----  
##                               carbidder  
##                               (1)      (2)      (3)      (4)      (5)      (6)  
## -----  
## all_stock      -0.023***   -0.022***   0.011***   0.015***   0.011***   0.013***  
##                t = -5.581 t = -5.322 t = 2.614 t = 3.545 t = 2.796 t = 3.244  
##  
## deal_value                -0.00000***  
##                t = -3.813  
##  
## bidder_size                0.000  
##                t = 0.201  
##  
## log(deal_value)                0.018***           0.006***  
##                t = 9.712           t = 5.072  
##  
## log(bidder_size)                -0.022***           -0.012***  
##                t = -13.275          t = -10.315  
##  
## bidder_lev                0.032***           0.002           0.017**  
##                t = 2.954           t = 0.175           t = 1.967  
##  
## bidder_fcf                -0.007           -0.001           -0.001  
##                t = -0.653          t = -0.114          t = -0.092  
##  
## horz                -0.001           -0.001           -0.003  
##                t = -0.318          t = -0.188          t = -0.911  
##  
## hostile                0.002           0.002  
##                t = 0.107           t = 0.118  
##  
## public                -0.018***   -0.013***  
##                t = -4.924 t = -3.092  
##  
## all_stock:public                -0.034***   -0.041***
```

```

##                                     t = -5.646 t = -6.856
##
## Constant          0.00004      -0.002    0.018***    0.226***    0.018***    0.141***
##                   t = 0.016   t = -0.616   t = 6.700 t = 13.450   t = 7.165   t = 11.879
##
## -----
## Observations      2,385        2,385        2,769        2,769        5,154        5,154
## R2                0.013        0.024        0.002        0.066        0.028        0.053
## Adjusted R2       0.012        0.021        0.002        0.064        0.027        0.051
## =====
## Note:                                     *p<0.1; **p<0.05; ***p<0.01

```